

Water Through Rock Lab

BUILD • Science • 40 minutes

I can follow or direct a fixed water-passage method, record what reaches the collector and make a careful comparison of supplied samples.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

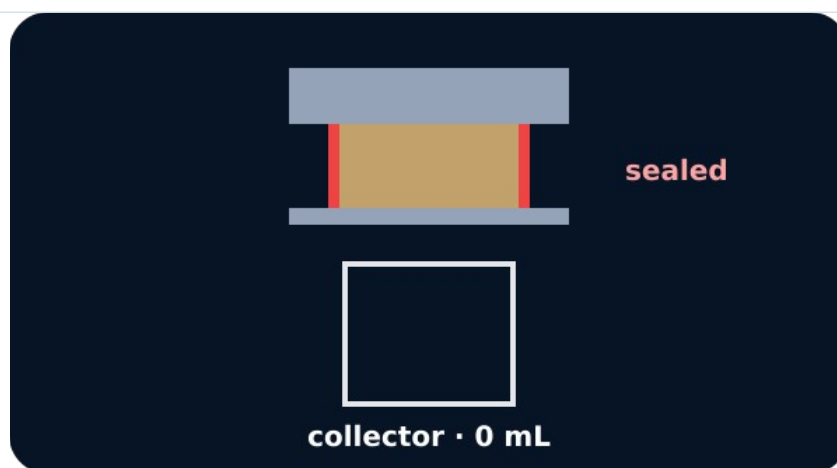
Word	Meaning
permeability	how water passes through
volume	amount of liquid
valid	the method worked as stated

First idea

Water appears in the tray, but the collar leaked around the sample edge. What should the lab team record?

- A. Count all tray water as passed through the rock.
- B. Mark the run invalid because water bypassed the sample; stop and use the repaired or fixed-data route.
- C. Guess how much probably went through and keep the result.

My first idea and reason:



A sealed sample above a collector

We do • choose, explain, check

Sample A collects 0.5 mL and Sample B collects 3.0 mL after the same 5 mL and 120-second method. Which comparison is supported?

- A. More water passed through supplied Sample B under this method.
- B. Sample A is waterproof forever.
- C. Every rock like B always passes exactly 3.0 mL.

My choice: _____ Evidence or reason:

Water-passage lab console

Route each lab record to VALID A, VALID B, METHOD FAULT or CAREFUL COMPARISON.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
VALID • SAMPLE A	
VALID • SAMPLE B	
METHOD FAULT • DO NOT COMPARE	
CAREFUL COMPARISON	

Activity cards

1 • A • sealed run	2 • A • surface note
5 mL added; 120 seconds; seal held; 0.5 mL collected below.	A small bead remained on top and the collector held 0.5 mL; no edge leak was seen.
3 • B • sealed run	4 • B • underside note
5 mL added; 120 seconds; seal held; 3.0 mL collected below.	Drops formed under the prepared sample inside the sealed area and entered the collector.

5 • Loose collar

Water ran down the outside edge into the tray; the collector reading cannot be separated from the leak.

6 • Timer missed

The team cannot say whether the run lasted 60, 120 or 180 seconds.

7 • Bounded result

Under the fixed method, more water passed through supplied Sample B than supplied Sample A.

8 • Time caution

Little water in 120 seconds does not prove that a sample is waterproof for every time and use.

Use the frozen records to say what passed through, which run was valid and what the result does not prove.

Your lesson record

Follow or direct the fixed water-passage lab for supplied Samples A and B, or analyse the fixed equivalent records, then give a bounded comparison.

Model helps us see

Check in order: seal held → 5 mL added → 120 s timed → collector read in mL → claim names supplied samples and method.

Model does not show

The lab uses a short classroom test on prepared samples. Natural variation, tiny cracks, sample thickness and sealing can affect results; the test is not a full building-performance check.

Supported

Sequence the fixed method, place A and B result cards and complete an oral, pointed or scribed comparison.

Use: A collected ___ mL. B collected ___ mL. More water passed through ___ in this test.

Standard

Record both valid results, reject one fault record and explain the sample comparison with a time limit.

Check seal, volume, time, collector and claim boundary in order.

Stretch

Evaluate how thickness, seal or natural variation could affect the comparison and propose one control to investigate in W13.

Do not rewrite the observed values; improve the future method.

Evidence target

Method checklist, surface/underside observation, collector values with mL, valid/fault decision, bounded comparison and one method-control question for W13.

Exit, Audience and evidence • W11B

Supported: Which supplied sample had more water reach the collector — A or B?

Standard: Give the comparison and explain why it does not mean waterproof forever.

Stretch: Name one method control that needs closer planning before a larger fair test.

What I said, and what it changed

Next I will _____.

Adult witness note

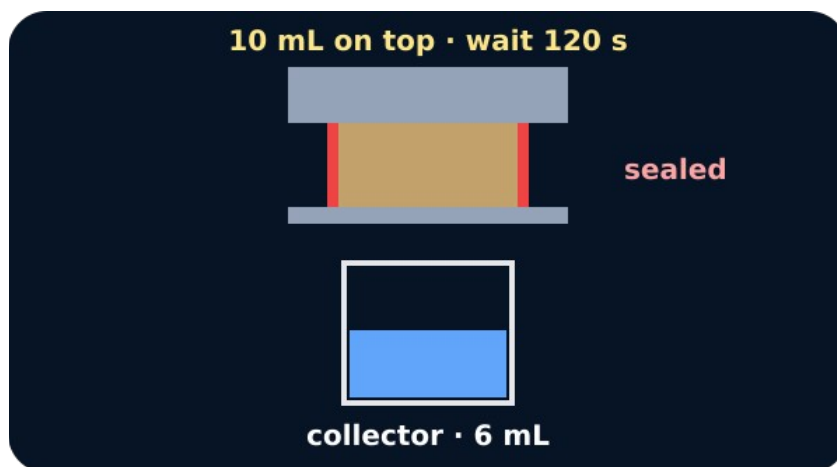
What was observed, and with what level of independence?

My evidence and explanation

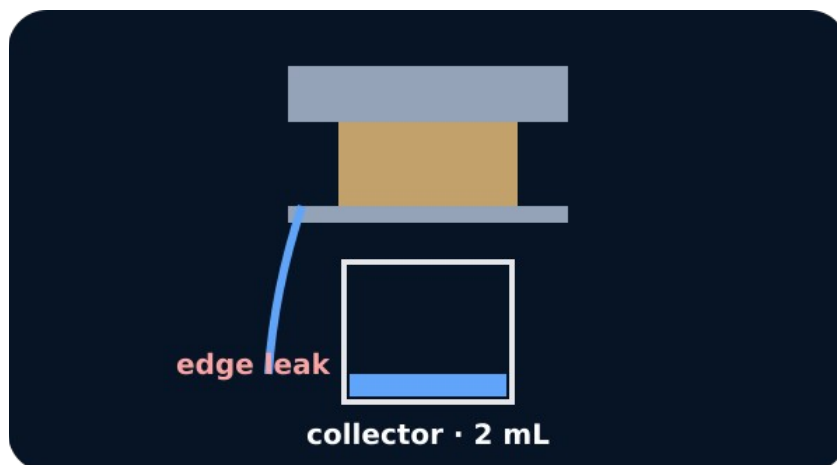
What does this evidence not show?

Visual reference cards

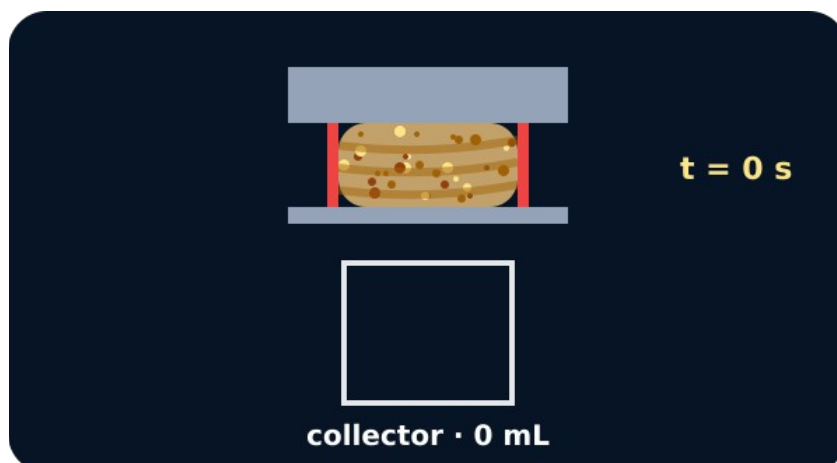
These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



Water collecting below the sample



An edge leak spoiling a result



Sealed water bench with collector

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
